COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit – 1 : Session – 1 : SLO – 1 | Functional units of a computer
Name: JATANPREET SINGH VIRDI JATANPREET SINGH VIRDI | Reg. No: RA2511003011374 | Solved Worksheet

I. Multiple Choice Questions

• 1. C — To design better programs and optimize performance
• 2. B — Operating System
• 3. B — Instruction Set Architecture
• 4. B — Level 5 – High-Level Language Level
• 5. C — Von Neumann bottleneck

II. True or False

• 1. TRUE — The system software level (OS) handles memory protection and process synchronization.
• 2. FALSE — Cache memory is faster (and smaller) than RAM.
• 3. FALSE — Machine-level programs execute directly; compilers are needed for high-level languages.
• 4. TRUE — Harvard architecture uses separate buses for instructions and data.
• 5. FALSE — Control units can be hardwired or microprogrammed.

III. Scenario-Based Activity

• Suggested change: Move from pure von Neumann architecture to a Harvard / modified-Harvard architecture with separate instruction and data buses (and add cache memory plus more RAM).
• Why: In von Neumann architecture a single shared bus forces instruction fetches and data transfers to take turns, creating the von Neumann bottleneck. Separate buses let the CPU fetch instructions and read/write data simultaneously, and cache reduces trips to the slow main memory. Together these remove the memory lags and raise throughput without changing the programs themselves.